

ORIGINAL RESEARCH

GOMT: Multispectral video tracking based on genetic optimization and multi-features integration

Kun Qian¹  | Peng Chen¹ | Dong Zhao²

¹School of Artificial Intelligence and Computer Science, Jiangnan University, Wuxi, Jiangsu, P. R. China

²School of Electronic Information Engineering, Wuxi University, Wuxi, Jiangsu, P. R. China

Correspondence

Kun Qian, School of Artificial Intelligence and Computer Science, Jiangnan University, No.1800, Lihu Road, 214122, Wuxi Jiangsu, Jiangsu Province, P. R. China.
Email: kqian@jiangnan.edu.cn

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Abstract

Traditional visual tracking algorithms on RGB cannot effectively distinguish an object from background with similar colour feature, resulting in the migration problem in the tracking process. In order to solve this problem, a tracking algorithm with spectral information is proposed by using band selection, information fusion, and deep features. Firstly, the correlation analysis of 16 spectral band information is carried out, and genetic optimization method is introduced to obtain two bands with low correlation coefficient and abundant information. Moreover, under the framework of kernel correlation filtering tracking, guided filtering is adopted to fuse the information referring to the target region of the two bands. Besides, the feature maps are generated by histogram of gradient and pretrained visual geometry group network. Finally, target is detected via finding the maximum value of a strong response. The proposed genetic optimization based multifeature tracker is compared with multiple tracking methods including correlation filtering and deep learning. Experimental results with multiple groups of spectral videos and corresponding RGB videos demonstrate that the genetic optimization based multifeature tracker method achieves good results in subjective vision and objective evaluation.

1 | INTRODUCTION

Multispectral image (MSI) plays an important role in remote sensing, for the reason that it provides tens of contiguous spectral bands [18]. With rich spectral information, multispectral images (MSIs) have been widely used in many applications, including target detection [7, 44] and classification [6, 42]. Recently, object tracking on MSIs has been popular because of that low cost multispectral video (MSV) cameras become available. However, some problems need to be solved in the tracking task [26, 36] urgently.

Generally, there are two types of visual tracking methods, involving discriminative correlation filter (DCF) based methods [3, 9–12, 14, 39, 43] and generative deep learning (DL) based methods [2, 15, 17, 19, 25, 29, 30, 45]. The DCF trains filters [20, 23] efficiently in the frequency domain via fast Fourier transform (FFT). It aims to obtain a correlation filter that denotes the inherent difference between the target and its neighbourhood. Bolme et al. proposed the fast minimum output sum of squared error (MOSSE) tracker [3], which

uses greyscale features and achieves a superhigh speed. Yet, the MOSSE tracker is sensitive to the complex environment, for the reason that only raw grey feature is utilized for target matching. Moreover, some improvements use other features, such as colour name features [5], histogram of gradient (HOG) features [10], and DL features [25]. The kernelized correlation filter (KCF) method [10] exploits the advantage of multichannel HOG features with the kernel trick. The HOG based feature extraction method helps to improve the tracking accuracy referring to targets with salient contour. Further, Martin et al. proposed a spatially regularized discriminative correlation filters (SRDCF) [4], which efficiently alleviates the boundary effect. Wang et al. presented a multi-cue correlation based tracker (MCCT) [31] with multifeature of HOG and visual geometry group (VGG). Deep features automatically capture the intrinsic properties of objects in different scenes. These improved DCF trackers attempt to solve the occlusion problem [45] of the original MOSSE method. Nevertheless, these discriminative methods are also sensitive to the false target with colour similarity.

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In addition, the DL tracking focuses on the extraction of deep features, such as the deep learning tracker (DLT) [30] and siamese network tracking [2, 17, 28, 32] methods. Generally, it adopts the mode of offline training with online tracking, which updates the parameter and extracts inherent features of the object. The DLT tracker first applies the deep network to target tracking, but it is limited by the size of input images needs to be fixed. Several works, such as the fully convolutional neural networks (CNN) tracking [29] method and the siamese region proposal network [13] tracking method, have combined DL with the DCF based tracking framework. Instead of obtaining hand-crafted features, the DCF trackers adopted features automatically learnt via CNN, which improves the robustness of the tracking significantly. Hang et al. proposed the lightweight CNN based tracker [37] without training process, which makes the DL based tracking algorithms achieve real-time performance. However, the accuracy of the above visual tracking methods is easily affected by similar colour in the background. Besides, independent graphics processing unit (GPU) is needed to computing deep features in most cases.

In order to improve the tracking performance, this paper intends to use the MSV to achieve the region of interest state estimation [28, 36]. The MSVs contain large amounts of information referring to 16 bands in the range of 470–620 nm. Therefore, feature dimension reduction or MSIs features can be utilized to cope with the multispectral data. In recent years, some researchers have tried the tracking methods based on MSIs and obtained effective tracking results [28, 36]. Kun et al. utilized the lightweight CNN to obtain the original spectral information [28]. Specially, a multidimensional filter was selected to extract multidimensional spectral features. Yet, the redundant information of MSIs results in the insufficient ability of feature representation. Xiong et al. integrated the spectral information with 3D HOG feature via the DCF tracker [36]. The spectral feature improves the performance of the tracker, greatly. Furthermore, Zhang et al. [41] realized a multi-feature integration based hyperspectral videos tracking (MFIHVT), which combines VGG features with the correlation filtering. However, dimension reduction method based on principal component analysis (PCA) [22] easily destroys the intrinsic structure information of MSIs. To our knowledge, some intelligent optimization algorithms were introduced into the band selection of MSIs [25], which preserved most of the original information, and improved the execution efficiency.

In this paper, the genetic optimization [42] and multifeature based tracker (GOMT) is proposed for MSVs processing. We first utilize the optimization algorithm to select a band group with high information entropy. In the tracking step, the KCF is adopted to distinguish targets from neighbouring environment. The multifeature model via the 3D HOG and VGG network helps to obtain a strong response. The novelty and advantages of our work are summarized as follows:

(1) The influence of the colour similarity is reduced via the spectral properties of multispectral videos effectively.

- (2) Feature extraction is achieved based on the band selection of genetic optimization, the guided filtering fusion, and the deep VGG.
- (3) The experiment designed over multispectral videos and its corresponding RGB videos, demonstrates that the proposed method has good performance.

The remainder of this paper is organized as follows. In Section 2, the KCF method is described briefly. Then, we introduce the genetic optimization for band selection, and the guide filtering fusion for spectral information. In Section 3, the experimental results are presented to verify the performance of the proposed method on MSIs. The GOMT is also compared with the existing methods on RGB and greyscale videos of the same scene. Finally, conclusions are drawn in Section 4.

2 | TRACKING ON MULTISPECTRAL VIDEOS

In this section, the KCF tracker [10] is briefly described, and the component of feature extraction is given for the MSIs.

2.1 | The KCF tracker

The KCF algorithm, which utilizes the ridge regression model, aims to train a strong classifier. Besides, its objective function is formulated as

$$\min_{\mathbf{w}} ((\mathbf{w}^T \mathbf{x} - y)^2 + \lambda \|\mathbf{w}\|^2) \quad (1)$$

where λ is regularization parameter. $\mathbf{w}$ denotes the regression coefficient, and $\|\mathbf{w}\|^2$ is the square function, representing the two norm regularization. (x, y) represents the sample and its label value.

Given a coefficient $\mathbf{a}$ in high-dimensional feature space, $\mathbf{w} = \mathbf{a}^T \varphi(\mathbf{x})$ represents the mapping operation, and $\varphi(\cdot)$ is the mapping function. With kernel trick, the new coefficient $\mathbf{a}$ is represented by

$$\begin{aligned} \mathbf{a} &= (\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{y} \\ \mathbf{F} \mathbf{a} &= ((\mathbf{F} \mathbf{K}^{\text{xx}}) + \lambda) \mathbf{F} \mathbf{y} \end{aligned} \quad (2)$$

where $\mathbf{y}$ is a prior in the current frame and $\mathbf{F}$ represents the FFT matrix. The kernel matrix $\mathbf{K}$ with $\mathbf{k}^{\text{xx}}$ denoting the first row is a circulant matrix, which can be decomposed into

$$\mathbf{K}(\mathbf{k}^{\text{xx}}) = \mathbf{F} \cdot \text{diag}(\mathbf{F} \mathbf{k}^{\text{xx}}) \cdot \mathbf{F}^H \quad (3)$$

where $\mathbf{F}^H$ is the Hermitian transpose of $\mathbf{F}$, and $\text{diag}(\cdot)$ is the diagonalisation function.

In Equation (4), k can be computed based on the Gaussian function, that is,

$$k^{\mathbf{x}} = \exp \left(-\frac{1}{\sigma^2} (\|\mathbf{x}\|^2 + \|\mathbf{z}\|^2) - 2F^{-1} \left(\sum_d (\mathbf{F}x_d) \otimes (\mathbf{F}z_d) \right) \right) \quad (4)$$

where x_d denotes the d th target representation of the multi-channel representation. Besides, $\otimes$ denotes the image convolution.

Subsequently, the tracking task is treated as a detection problem, which can be solved via obtaining a response map.

$$f(\mathbf{z}) = \mathbf{F}^{-1}((\mathbf{F}k^{\mathbf{x}}) \otimes (\mathbf{F}\mathbf{a})) \quad (5)$$

where $\mathbf{x}$ and $\mathbf{a}$ are updated before the current frame. In the current frame, the image patch $\mathbf{z}$ at location $\mathbf{x}$ denotes the testing base sample. The response $f(\mathbf{z})$ can be intuitively described as a linear combination of the neighbouring kernel value $k^{\mathbf{x}}$ with the weighted coefficient $\mathbf{a}$.

2.2 | Band selection and information fusion

In order to reduce the complexity of the tracking process and the redundancy of the MSIs, whose intrinsic feature needs to be obtained via feature dimension reduction methods, including feature selection [35] and feature extraction [39]. The former refers to selecting features with good classification performance, i.e. band selection of MSIs. Feature extraction aims to map features from the original space to a new low-dimensional space, such as PCA, linear discriminant analysis and singular value decomposition. Yet, it needs to give new definition of feature attributes. Therefore, this paper utilizes genetic algorithm [42] to retain most of the original information.

2.2.1 | Band selection via genetic optimization

In general, the correlation between retained bands should not be too high. The diagram of correlation coefficient and the operation of band selection are shown in Figure 1.

As can be seen from Figure 1, the matrix diagram of the correlation coefficient indicates that the initial images can be divided into two groups, i.e. (1–10) and (11–16). Moreover, the correlation coefficient of intra-group images is high, and the inter-group correlation is low inversely. Therefore, the redundancy of original information should be reduced.

In view of the large amount of multispectral video data, the calculation of information entropy between bands is also time-consuming. Based on the above analysis, genetic optimization algorithm [1, 23] is introduced to quickly obtain two bands with a maximum entropy value. Given the initial image $\mathbf{D} =$

$[d_1; d_2; \dots; d_{16}] \in R^{m \times n \times 16}$, the joint information entropy $\mathbf{H}$ of band d_i and d_j can be expressed as

$$\mathbf{H}(d_i, d_j) = - \sum_{I,J=0}^{255} \mathbf{P}_{I,J} \log_2 \mathbf{P}_{I,J} \quad (6)$$

where, $\mathbf{P}_{I,J}$ is the joint probability between pixel value I of band d_i and pixel value J of band d_j .

According to the above analysis, the joint entropy can be seen as the fitness function of the discrete genetic algorithm. Besides, the band number group is used as the chromosome in the genetic algorithm. In order to obtain two suitable bands, selection, crossover, and mutation are implemented iteratively, which is shown in Figure 1. Experimental results show that the optimal option referring to the initial image of toy1 is 8 and 15, which verifies the conclusion about the above correlation analysis.

To sum up, two bands in each group are selected based on the joint entropy, which takes into account the information reduction and the execution efficiency. That is to say that, increasing the number of bands for the given spectral data will result in information redundancy.

2.2.2 | Information fusion via guided filtering

For obtaining a strong target representation, information fusion is carried out on the optimal bands. Guided filtering Fusion (GFF) [16] is a weighted fusion method, which only needs to obtain two scales of images and takes the saliency of pixels and spatial continuity into consideration. Given two bands $\mathbf{C}_n \{n = i, j\}$, the two scales can be obtained by mean filter,

$$\begin{aligned} \mathbf{B}_n &= \mathbf{C}_n * \mathbf{Z} \\ \mathbf{D}_n &= \mathbf{C}_n - \mathbf{B}_n \end{aligned} \quad (7)$$

where $\mathbf{B}_n$ and $\mathbf{D}_n$ represent the base layer and detail layer of band $\mathbf{C}_n$, respectively. The larger the size of filter $\mathbf{Z}$, the more fuzzy the base layer is, and the more information the detail layer contains.

Subsequently, the mapping coefficients $\mathbf{W}_n^{\mathbf{B}}$ and $\mathbf{W}_n^{\mathbf{D}}$ referring to the base layer and detail layer can be computed via the guided filtering [8],

$$\begin{aligned} \mathbf{W}_n^{\mathbf{B}} &= G_{r_1, \varepsilon_1}(\mathbf{P}_n, \mathbf{C}_n) \\ \mathbf{W}_n^{\mathbf{D}} &= G_{r_2, \varepsilon_2}(\mathbf{P}_n, \mathbf{C}_n) \end{aligned} \quad (8)$$

where $G(\cdot)$ denotes the guiding filter function with corresponding parameters r_1 , ε_1 , r_2 , and ε_2 . The base layer and the detail layer correspond to larger and smaller parameters, respectively. In addition, the mapping coefficient $\mathbf{P}_n$, which only considers saliency, is obtained by Laplacian filter L , Gaussian low-pass

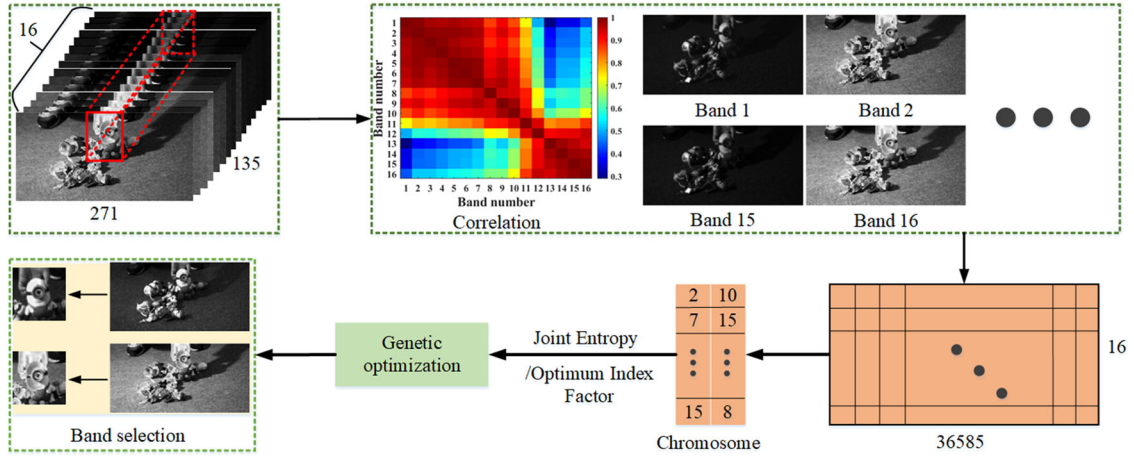


FIGURE 1 Band selection for the initial toy1 image.

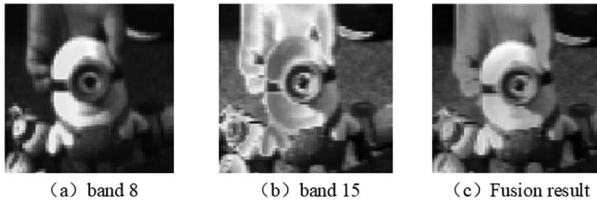


FIGURE 2 GFF method on toy1 (Entropy=6.48, 7.27, and 7.29. Average gradient=6.7, 7.9, and 8.2)

filter g , and maximum principle from the original band $\mathbf{D}_n$,

$$\mathbf{H}_n = \mathbf{C}_n * L \quad (9)$$

$$\mathbf{S}_n = |\mathbf{H}_n| * g$$

$$\mathbf{P}_n^k = \begin{cases} 1 & \text{if } \mathbf{S}_n^k = \max(\mathbf{S}_i^k, \mathbf{S}_j^k) \\ 0 & \text{else} \end{cases} \quad (10)$$

where $\mathbf{S}_n^k$ represents the saliency value of the k th pixel.

Accordingly, the fusion result $\mathbf{F}_{\text{GFF}}$ of GFF method is expressed as

$$\mathbf{F}_{\text{GFF}} = \sum_{n=i,j} (\mathbf{W}_n^B \mathbf{B}_n + \mathbf{W}_n^D \mathbf{D}_n) \quad (11)$$

According to the evaluation criterion of image quality in paper [16], experimental fusion result of band 8 and 15 (i.e., $i = 8, j = 15$) is shown in Figure 2.

Figure 2 indicates that fusion result is brighter than that of Figure 2a in image intensity, and is clear than that of band 15 without GFF method in details. At the same time, the information entropy and average gradient increase with fusion, explaining that the fusion image Figure 2c contains more information and details. Specifically, image entropy is described as the average image intensity or information. Larger entropy value indicates that the image quality is better. The average gradient is expressed as $(\sum_{i=1}^M \sum_{j=1}^N \sqrt{(\partial f / \partial x)^2 / 2 + (\partial f / \partial y)^2 / 2}) / (M \times N)$, where $M \times N$ is the image size, $\partial f / \partial x$ and $\partial f / \partial y$ rep-

resent the horizontal and vertical gradients, respectively. Thus, the average gradient reflects the sharpness and texture change of the image and the higher the average gradient value, the clearer the image.

2.3 | Multiple features for KCF

Each feature has its particular advantages in target matching. Specially, the HOG feature can deal with target deformation [10], yet the well-designed feature cannot adapt to all kinds of complex environments. To make up for this deficiency, deep features can be utilized to extract appropriate features for various scenarios.

In this paper, the HOG feature is utilized as shallow feature, and the deep features uses outputs of the conv(3-4), conv(4-4), and conv(5-4) layers in VGG network [31]. These outputs of layers have been proved to be effective in video tracking. Moreover, several weak responses $R_i, i = 1, 2, 3, 4$ are obtained with these multiple features, which can adapt to complex background. Specially, a response map is realized by using a filter to correlate an image, and its peak value represents a high degree of similarity between the current frame target and the template frame target. Therefore, the adaptive weighting coefficient β_i is represented as

$$\beta_i = RC_i / RH_i \quad (12)$$

where RC_i represents the i th peak response value of current frame. Besides, RH_i is the i th peak response value of all historical frames, which can be updated by

$$RH_i = \begin{cases} \mu \times RH_i + (1 - \mu) \times RC_i & \text{if } RC_i < RH_i \\ RC_i & \text{else} \end{cases} \quad (13)$$

Subsequently, the strong response R_s is formulated as

$$R_s = \left(\sum_{i=1}^4 \beta_i \cdot R_i \right) / 4 \quad (14)$$

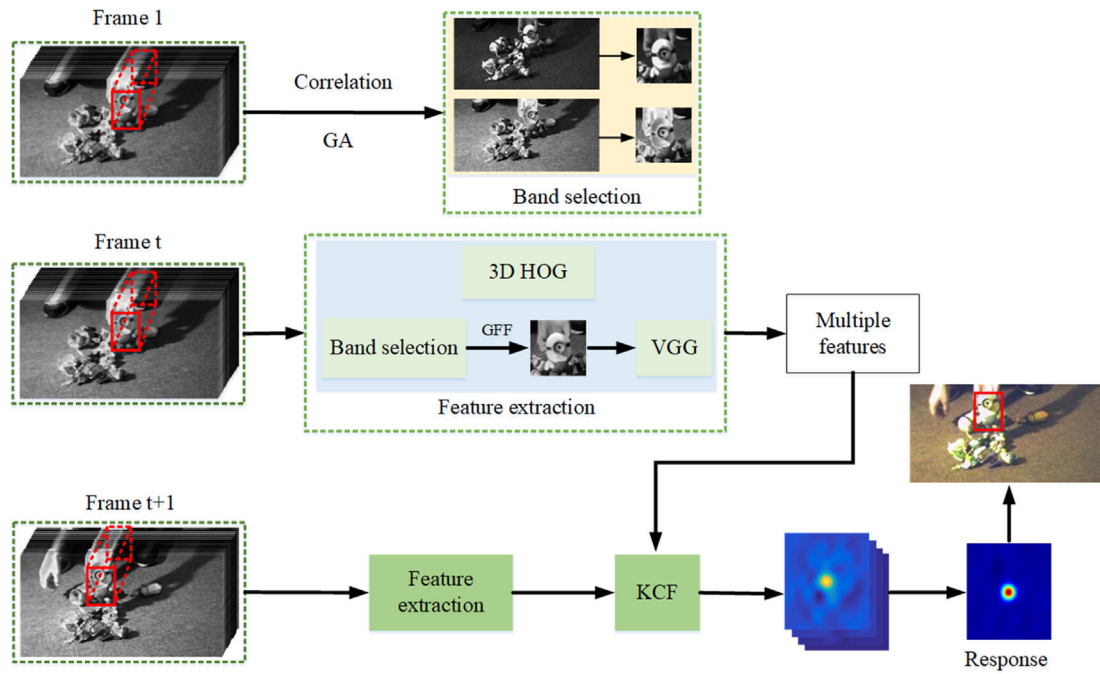


FIGURE 3 Flowchart of the proposed algorithm

Figure 3 shows the flowchart of the proposed algorithm, it takes advantage of the well-designed feature and the inherent representation. Moreover, the multiple features can be seen as several independent modules, referring to a strong response in KCF tracker.

In Figure 3, the content of the first row shows that the proposed tracker uses the first image to deal with band selection process. Specifically, for different videos with complex background, we only use the initial image to determine the number of the two bands, taking rich information and execution efficiency into consideration. Therefore, the same number, referring to the specific band, is utilized to pre-process all subsequent image frames.

3 | EXPERIMENTAL ANALYSIS

3.1 | Experimental data and parameter

Experiments are performed on types of videos with different challenging factors, including background clutters (BC), occlusion (OCC), out-plane rotation (OPR), in-plane rotation (IPR), deformation (DEF), low resolution (LR), and fast motion (FM), which are summarized in Table 1. The Nikon D600 camera and a Photonfocus multispectral camera were put side by side when capturing the videos with same scene. The multispectral camera captured frames of 16 bands with active range of 460–630nm. The greyscale video is formed by band image at 490nm of the MSI sequences. Therefore, the greyscale sequences are the same as the MSI sequences in the size, the frame number, the content, and the target location.

TABLE 1 Overview of test MSIs. (The total number of frames is 3903)

Video	Frame	Difficulty	Description
<i>book</i>	601	IPR/DEF/OPR	A book
<i>bus</i>	131	LR/BC/FM	A bus on the road
<i>coin</i>	149	BC	A coin with background clutter
<i>coke</i>	731	BC/FM/IPR/OPR	A coke bottle with similar background colour
<i>face2</i>	1111	OCC/IPR/OPR	A face under the circumstances of partial occlusion
<i>paper</i>	278	BC/IPR	A paper with similar background colour and appearance change
<i>Rubik</i>	526	DEF/IPR/OPR	A Rubik with severe deformation and rotation
<i>toy1</i>	376	BC/OCC	A stuffed toy occluded by a plastic toy with colour similarity

The proposed genetic optimization and multi-features based multispectral tracking (GOMT) method was implemented in MATLAB2020a and ran at 2 frame per second on a PC with 32G RAM. To better analyse the performance of the tracker, we compared it with several state-of-the-art algorithms, including DL based trackers DLT [31] and MCCT [32], DCF based trackers MOSSE [3] and SRDCF [4], and the MFIHVT algorithm [41] for MSVs. Qualitative results of existing trackers are shown in Figures 4–11.

For convenience, we display the results on RGB and pseudo-colour images, as their scene are identical to MSIs'. For all algorithms, the parameter on the search scope is changed, in

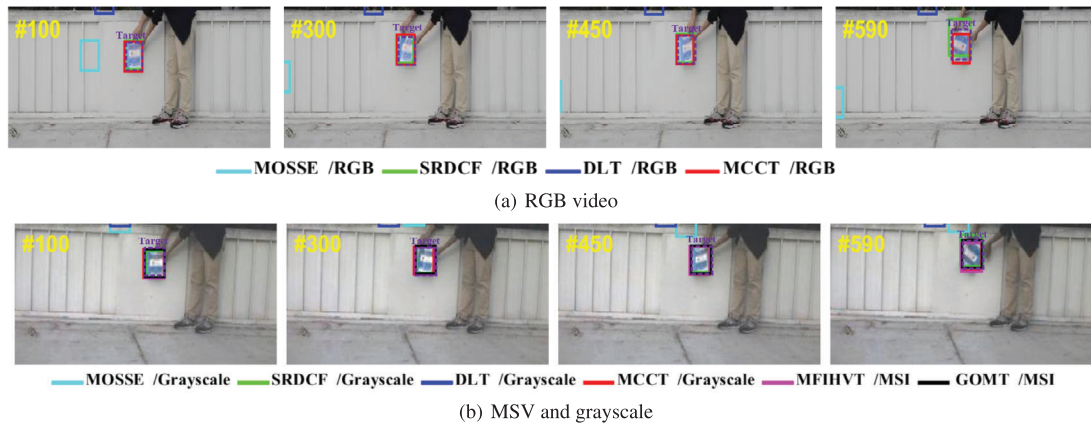


FIGURE 4 Qualitative results over *book* video (Best viewed on colour display)

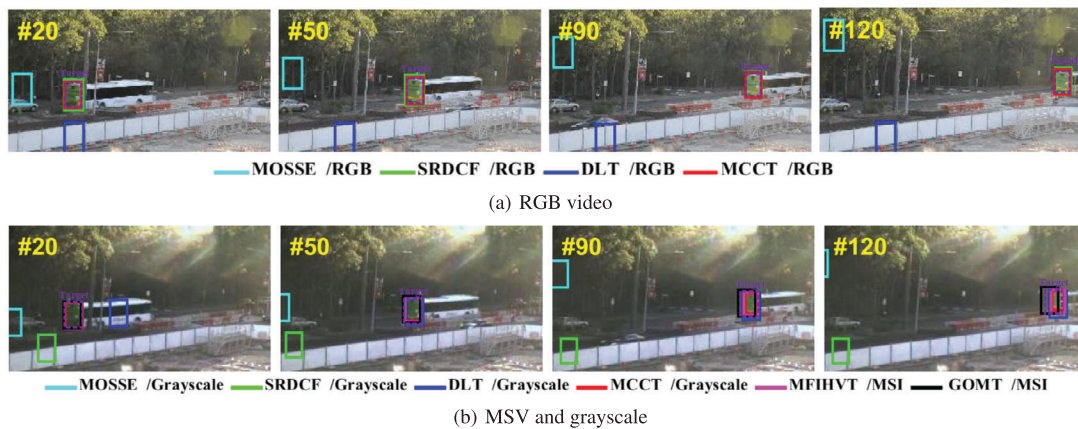


FIGURE 5 Qualitative results over *bus* video

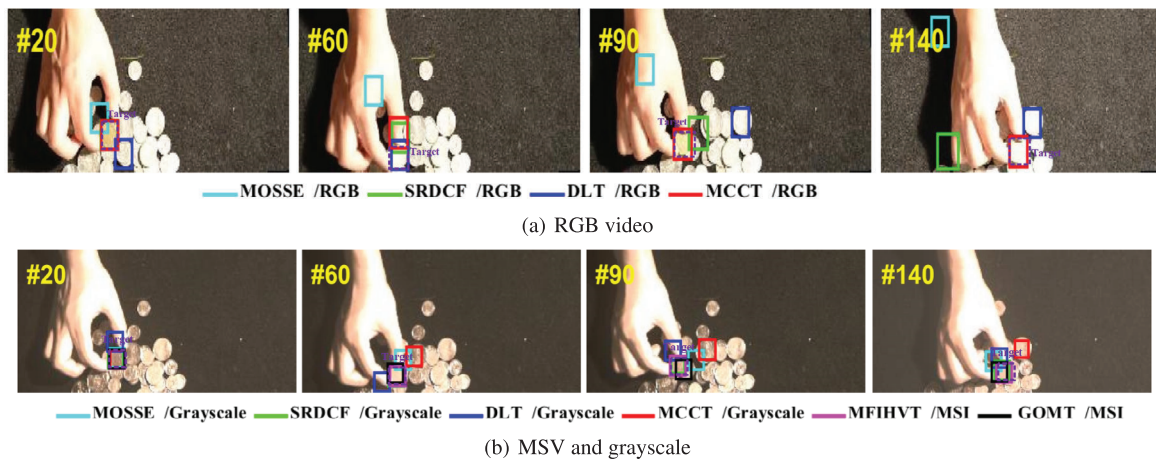
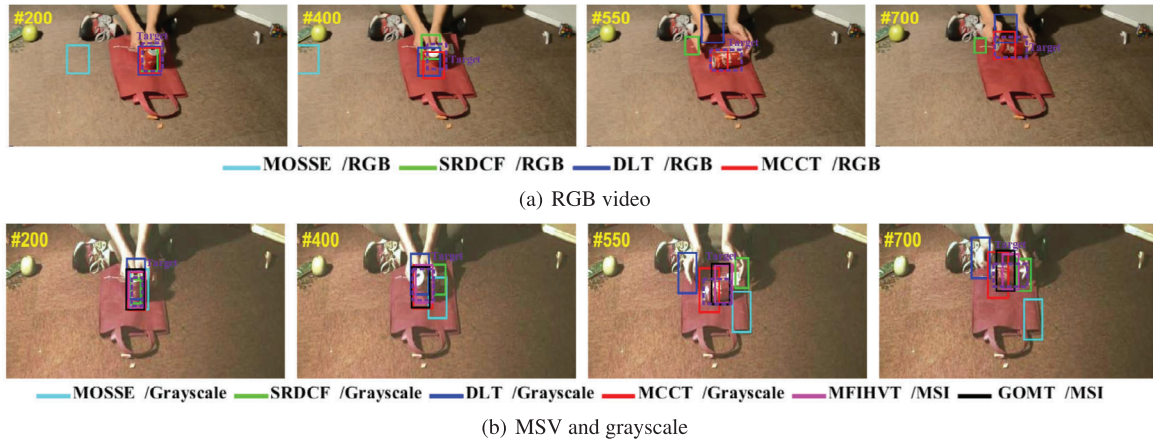
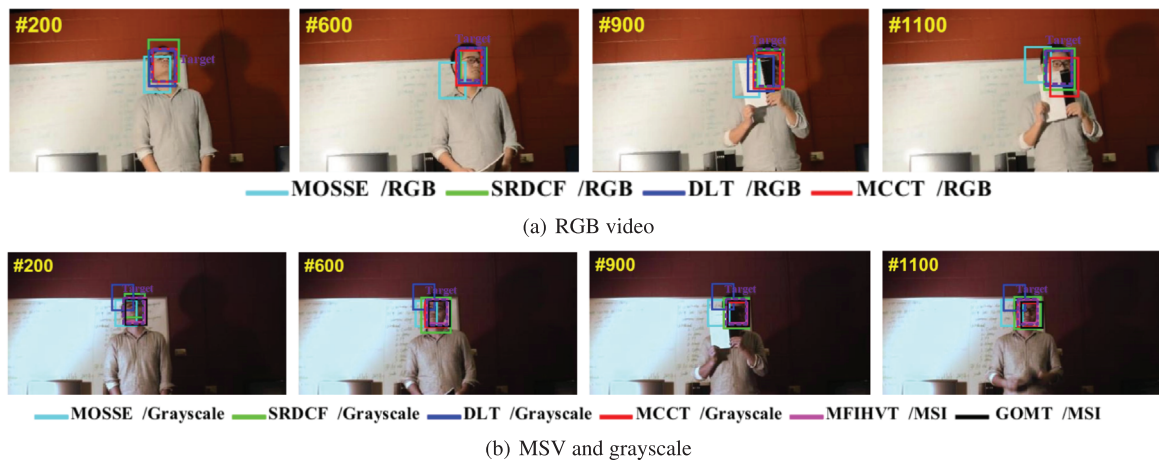
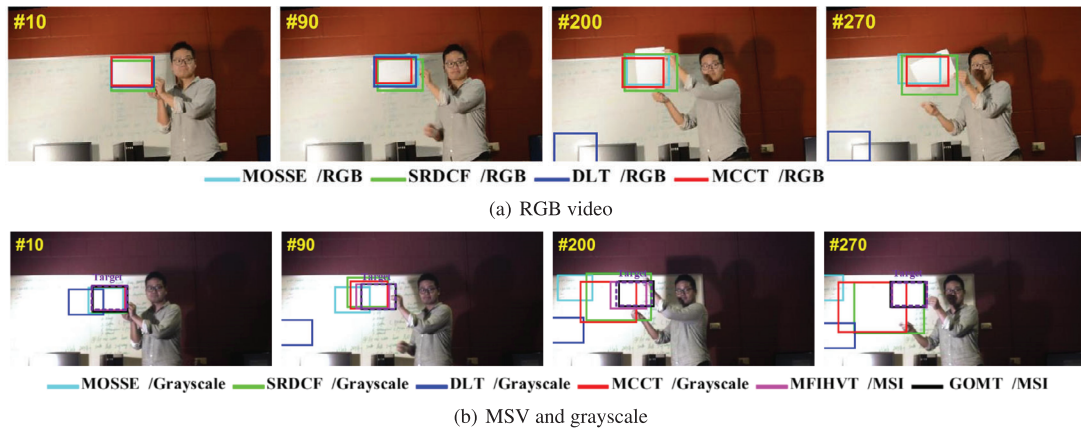


FIGURE 6 Qualitative results over *coin* video

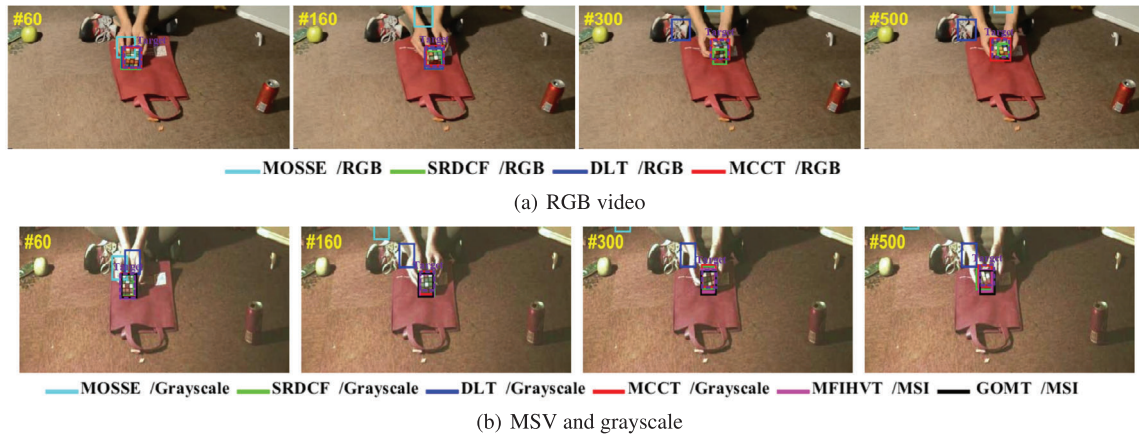
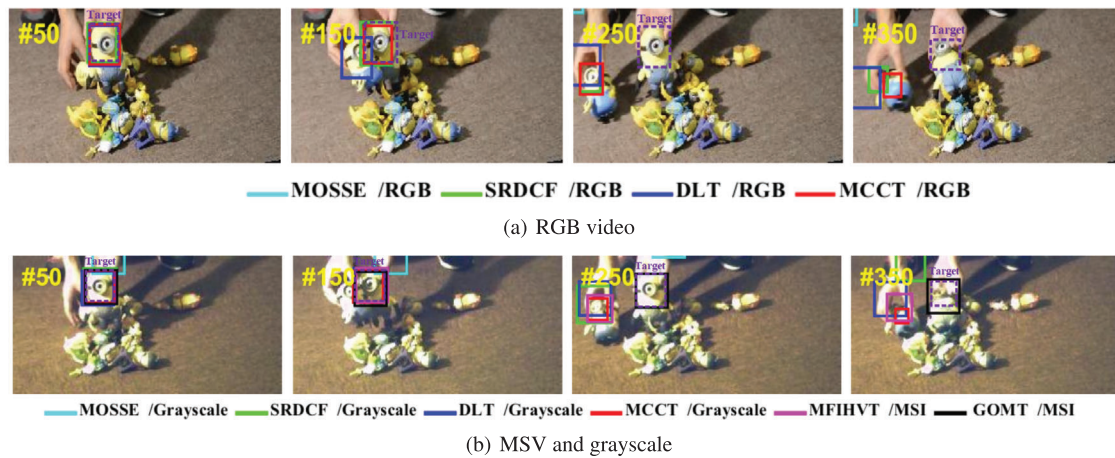
order to adapt to fast motion of the object. Besides, filter size in the part of GFF is appropriately adjusted in our tracker. In the part of band selection, we choose the spectrum group with the optimal index. For GFF method, the default filter size r_1

and blur degree ε_1 of the base layer are set as 45 and 0.3, the same parameter r_2 and ε_2 to the detail layer are 7 and 10^{-6} , respectively. We only change the filter size r_1 of the base layer to a fix smaller number 20, for the reason that the size of some

FIGURE 7 Qualitative results over *coke* videoFIGURE 8 Qualitative results over *face2* videoFIGURE 9 Qualitative results over *paper* video.

targets in the video is not large enough. Further, the parameter μ in formula (12) is set as 0.98, which means that the peak response value accepts some degree of online updating. In the

KCF-based tracking framework, the related parameters stay the same as those in the MCCT method [32], which is the basic algorithm of the proposed method.

FIGURE 10 Qualitative results over *Rubik* video.FIGURE 11 Qualitative results over *toy1* video.

3.2 | Qualitative comparison

3.2.1 | Background clutters and occlusion

Figures 6, 7, 9, and 11 show some screenshots of the tracking results in videos where the background and the target have similar colour. In the *coin* sequence, the colour of the coin is the same as its neighbourhood. In Figure 6a, the MOSSE method undergoes large drift in the entire sequence. The DLT and SRDCF track the target well at the beginning of the sequence (e.g., #20), but lose the target at #90 and #140. The tracking result of the MCCT method on the RGB video is more accurate. Utilizing the spectral information, the MFIHVT and our tracker perform well on the entire MSIs, which is shown in Figure 6b. In the *coke* sequence, the colour of the target bottle and the background are red. The MCCT and the GOMT stably track the object in the two types of videos. The MOSSE tracker drifts away from the target from the beginning to the end. The DLT and SRDCF approaches lock on parts of background when the hand appears in front of the bottle (e.g., #550 in Figure 7a,b). Moreover, Figure 9 shows that the visual tracking methods lose target at

200, for the reason that the greyscale of the paper and board are both white. Yet, the two MSI trackers obtain a convincing results from the beginning to the end.

The targets in Figures 8 and 11 face the problem of partial occlusion. In Figure 8, the face is occluded by a book at #900. In the RGB sequences, the DLT and the two improved KCF methods are able to re-detect the object when the target re-appears in the screen (e.g., #1100). Yet, the performance of the DLT tracker is influenced by the greyscale image, which can be seen in Figure 8b. In Figure 11, the stuffed target is partially occluded by another plastic toy with colour similarity. At #150, all trackers except the GOMT regard the false object as a true target, either in RGB videos or in MSVs.

3.2.2 | Rotation and deformation

Figures 4, 9, 10, and some other figures show the tracking results in videos where the target is subjected to deformation and rotation. For the *book* video, the MCCT, MFIHVT, and GOMT obtain stable tracking targets. For the #200 in *paper*

TABLE 2 Precision and FPS

Algorithm	Video type	<i>book</i>	<i>bus</i>	<i>coin</i>	<i>coke</i>	<i>face2</i>	<i>paper</i>	<i>Rubik</i>	<i>toy1</i>	Mean Precision (20px)	Mean FPS
MOSSE	RGB	7%	6%	16%	4%	19%	81%	10%	7%	18.8%	>1000 (CPU)
SRDCF	RGB	100%	100%	68%	60%	100%	86%	98%	56%	83.5%	11 (CPU)
DLT	RGB	13%	8%	47%	65%	86%	34%	29%	39%	40.1%	0.8/20 (CPU/GPU)
MCCT	RGB	100%	100%	100%	100%	93%	72%	100%	55%	90%	1 (CPU)
MOSSE	Greyscale	6%	5%	84%	50%	49%	27%	4%	10%	29.4%	>1000 (CPU)
SRDCF	Greyscale	100%	13%	100%	53%	100%	32%	98%	54%	68.8%	11 (CPU)
DLT	Greyscale	3%	82%	92%	46%	15%	4%	5%	42%	36.1%	0.6/10 (CPU/GPU)
MCCF	Greyscale	100%	100%	40%	91%	100%	31%	100%	53%	76.9%	1 (CPU)
MFIHVT	MSI	100%	100%	100%	64%	100%	97%	100%	54%	89.4%	1.5 (CPU)
GOMT	MSI	100%	100%	100%	97%	100%	100%	100%	100%	99.6%	2 (CPU)

greyscale sequence (Figure 9), the MOSSE, SRDCF, DLT, and the MCCT is sensitive to the rotation of the target. Nevertheless, the proposed method obtains a stable tracking target on the hyperspectral video with much better accuracy than the alternatives on the greyscale video and the RGB video. As shown in Figure 10a, the target in *Rubik* video undergoes deformation, resulting in tracking migration of DLT at #300 and #500. Furthermore, the DLT faces the same problem as in the greyscale image (at #160). However, the MSI based approaches handles deformation and rotation, thanks to the fact that it exploits the characteristic of hyperspectral features.

3.2.3 | Low resolution and fast motion

In Figure 5, the small target with fast motion contains little information. Thus, the DLT and the MOSSE methods lose targets at the beginning of the *bus* RGB video. For *bus* greyscale sequence, the MOSSE and SRDCF methods drift to the background after #20. In Figure 7, the MOSSE and SRDCF methods fail to track the bottle at #550 and #700, for the reason that the search scope is small. In addition, the MFIHVT and GOMT methods obtain a stable tracking target on the hyperspectral video.

3.3 | Quantitative comparison

Figure 12 shows the performance of the existing algorithms in terms of precision. The precision is treated as the ratio of successful frames, whose tracking output is within the given threshold (in pixels) from the groundtruth, measured by the centre distance between bounding boxes [33]. Table 2 shows the frame per second (FPS) and mean precision, referring to the threshold of 20 pixels.

As can be seen in Table 2 and Figure 12, the precision of the proposed algorithm on the MSV ranks the highest (0.996), which is followed by the MCCT (0.9) on RGB videos and MFIHVT (0.894) on MSVs. In most cases, tracking performance on RGB is higher than that on greyscale, which is because of that the RGB images contain more information. The MOSSE method only achieves the 18.8% mean precision. There maybe two main reasons why the precision of the MOSSE tracker is low. On the one hand, the MOSSE tracker does not consider background information, which makes the method be sensitive to the size of tracking box and the background clutter. Specifically, the target with large size seems to be detected well via the MOSSE, which is shown in Table 2 (The precision on the paper is 0.81). On the other hand, the purpose of this paper is to prove that MSVs can improve the performance of the tracking algorithm. Besides, most of the experimental videos contain an object with similar colour in its neighbourhood. Thus, visual trackers tend to have low tracking accuracy. The SRDCF method solves boundary effects via addition of spatial regularization into the DCF. Thus, it outperforms the MOSSE method by 3.8 on RGB videos. Finally, it is remarkable that the precision curve in Figure 12i demonstrate that only the GOMT effectively handle the occlusion of an object with colour similarity.

Afterwards, we briefly discuss the complexity of the proposed method for MSV tracking. The computational complexity of KCF [10] is as low as $O(n \log n)$, benefiting from FFT instead of expensive matrix algebra. As the GFF implementation needs to perform calculations over multiple windows, its computational cost is relatively higher. Besides, the GO can be performed in advance or per operation every 100 frames. Based on the aforementioned analysis, the entire computational complexity of the proposed method mainly depends on that of the VGG.

In short, the GO processing and the GFF fusion utilize spectral information. Besides, the deep VGG helps to obtain a strong response. The precision of the presented GOMT over

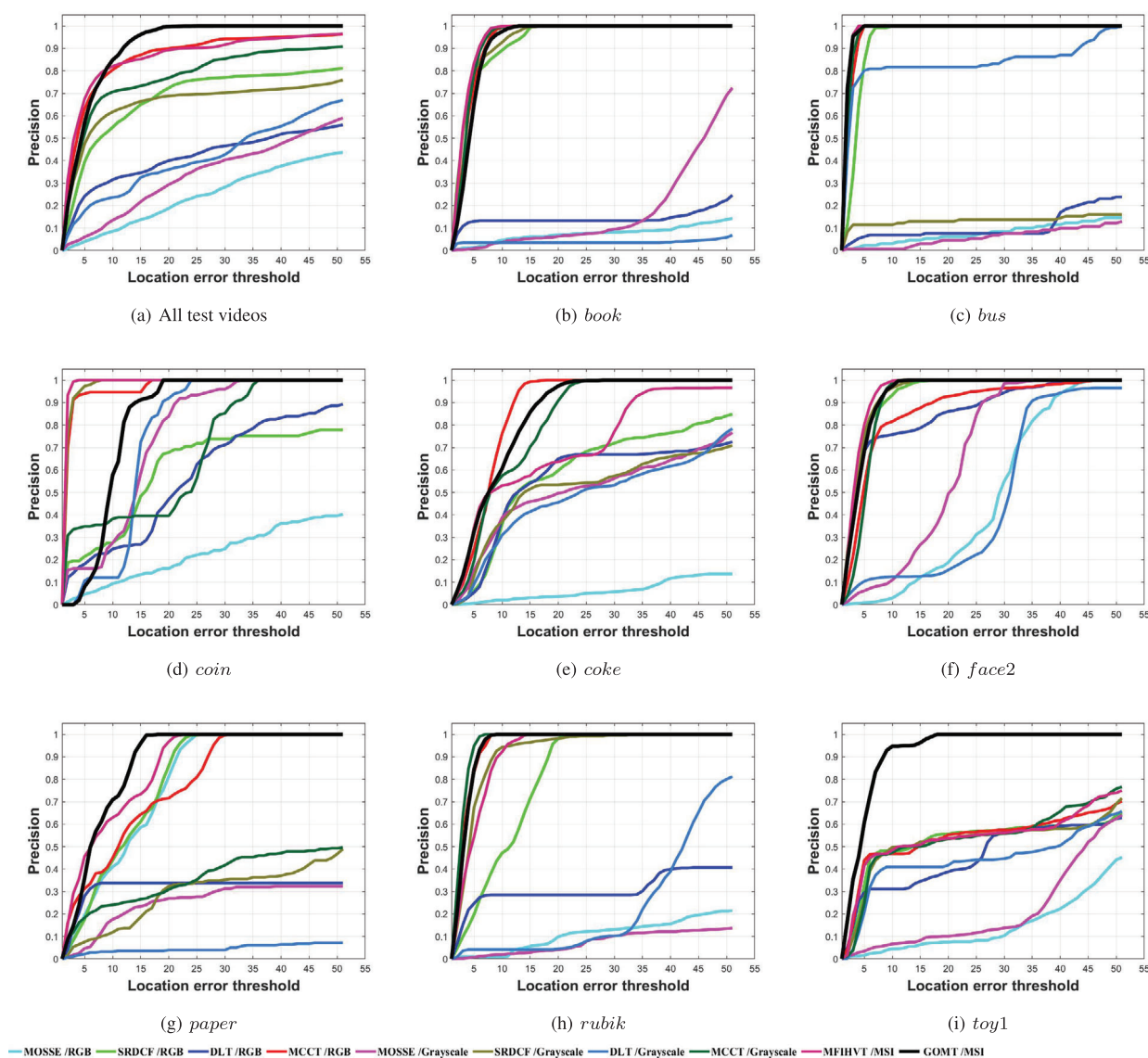


FIGURE 12 Precision plot of existing algorithms

all test videos is relatively satisfactory, proving that it performs better than related algorithms in the accuracy and robustness.

4 | CONCLUSIONS

In this paper, a genetic optimization based multifeature tracker is introduced for multispectral video tracking. The genetic algorithm and the guide filtering help to extend the discriminative correlation filter based method for multispectral data, and improve the efficiency of the tracking task. Besides, the VGG feature helps to obtain a strong response for target location. Most importantly, experimental results insist that the proposed tracker on MSVs, can effectively deal with the occlusion of an object with colour similarity to the target. This lays the foundation for developing future MSV tracking and identification algorithms.

AUTHOR CONTRIBUTIONS

Kun Qian: Conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, project administration, resources, software, supervision, validation, visualization, writing - original draft, writing - review and editing. Peng Chen: Data curation, resources, software, writing - review and editing. Dong Zhao: Conceptualization, formal analysis, funding acquisition, software, writing - review and editing.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available at <https://www.hsitracking.com/contest/>.

ORCID

Kun Qian  <https://orcid.org/0000-0003-2702-9500>

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